

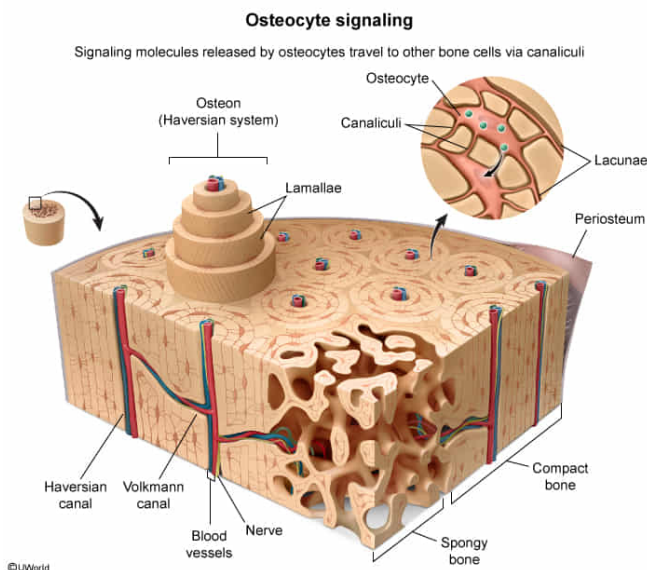
Which statement best explains how osteocytes regulate compact bone remodeling?

- ✔ ☒ A. Signaling molecules released by osteocytes travel to other bone cells via canaliculi. [37%]
- ☐ B. Osteocytes travel between lacunae to communicate with other bone cells. [14%]
- ☐ C. Osteocytes differentiate into cells capable of producing signaling molecules that regulate bone remodeling. [36%]
- ☐ D. Haversian canals provide a pathway for osteocyte travel in the bloodstream. [11%]

Correct

37% answered correctly

Explanation:



Bone is made of several types of cells surrounded by a firm matrix.

- Osteoprogenitor cells are stem cells that **differentiate** into another type of bone cell, the osteoblast.
- Osteoblasts participate in **bone remodeling** by secreting proteins that create the osteoid, or unmineralized bone matrix. In addition, osteoblasts promote the absorption of calcium and phosphate from the bloodstream and the incorporation of these molecules into the bone matrix. By building this matrix around themselves, osteoblasts eventually become surrounded and differentiate into osteocytes.

- Osteoclasts are large cells that participate in bone remodeling by breaking down (resorbing) old bone.
- Osteocytes are mature and mitotically inactive bone cells that maintain bone structure. These cells release signals to other bone cells to regulate **compact bone** remodeling.

Because osteocytes are surrounded by bone matrix and are located within spaces called lacunae in the **Haversian system**, they cannot travel to other cells during signaling. Rather, they release signaling molecules responsible for stimulating other bone cells into canaliculi, the small canals that connect cells within the bone matrix. **Signaling molecules** released by osteocytes **travel** through these small **canaliculi** until they reach their target cells (ie, other bone cells).

(Choices B and D) Osteocytes are trapped within lacunae of the firm bone matrix and thus *cannot* travel between lacunae or through Haversian (central) canals in the bloodstream to communicate with other bone cells.

(Choice C) Osteocytes are fully differentiated amitotic cells. They do not differentiate into cells capable of producing signaling molecules that regulate bone remodeling; rather, they release these signaling molecules themselves.

Educational objective:

Several types of bone cells (ie, osteoprogenitor cells, osteoblasts, osteocytes, osteoclasts) exist. Because osteocytes are confined to lacunae within a firm matrix in compact bone, signaling molecules released from osteocytes that regulate bone remodeling must travel to their target cells via canaliculi.

Time spent:0 sec

Subject:
Biology

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System:
Musculoskeletal System